TB Infectious Disease Modeling: Background and Literature Review

Contents

1 TB epidemiology and natural history **2**
- 1.1 Current global burden and regional distribution . . . 2
- 1.2 Twenty-year trend in global incidence . . . 2
- 1.3 Natural history in the absence of treatment . . . 3

2 Mathematical modeling approaches for TB **4**
- 2.1 Stochastic ODE models and uncertainty analysis . . . 5
- 2.2 Extended compartmental models with drug resistance and reinfection . . . 5
- 2.3 Individual-based and microsimulation models . . . 5
- 2.4 Risk-factor stratified models: nutrition and biomarkers . . . 6
- 2.5 Models of diagnostic interventions and the transmission curve . . . 7

3 Interactions with HIV **8**
- 3.1 Biological mechanisms and parameter modifications . . . 8
- 3.2 Modeling approaches for TB–HIV co-dynamics . . . 8
- 3.3 Impact of antiretroviral therapy on TB dynamics . . . 9
- 3.4 Implications for model design . . . 9

4 Economic evaluation approaches for TB **9**

5 The SELT(R) model used in this project **10**
- 5.1 Compartments and biological rationale . . . 10
- 5.2 Force of infection and the transmission curve . . . 11
- 5.3 Basic reproduction number . . . 11

1 TB epidemiology and natural history

Tuberculosis (TB) is caused by *Mycobacterium tuberculosis* and transmitted primarily via the airborne route. Transmission typically requires prolonged contact, and infectiousness varies by disease stage. The majority of infected individuals do not develop immediate symptomatic disease; instead they enter a latent state with an elevated risk of future progression [Vynnycky and Fine, 1997]. A commonly cited rule of thumb is that roughly 5–10% of infected individuals progress to active disease within the first few years, with a smaller but persistent risk of reactivation thereafter [Vynnycky and Fine, 1997, World Health Organization, 2025].

Global burden remains substantial. The WHO Global Tuberculosis Report 2025 documents ongoing high incidence in multiple regions and highlights the role of HIV co-infection in disease progression and mortality [World Health Organization, 2025]. The latent TB reservoir is large: a recent re-estimation suggests roughly one quarter of the global population has latent infection [Houben and Dodd, 2016].

1.1 Current global burden and regional distribution

In 2024, an estimated 10.7 million people fell ill with TB worldwide, equivalent to 131 new cases per 100,000 population [World Health Organization, 2025]. Eight countries accounted for two-thirds of global cases: India (25%), Indonesia (10%), the Philippines (6.8%), China (6.5%), Pakistan (6.3%), Nigeria (4.8%), Democratic Republic of the Congo (3.9%), and Bangladesh (3.6%).

Table 1 shows the most recent cross-sectional estimates by WHO region. The South-East Asia Region (SEAR) and the African Region carry the highest absolute and per-capita burden, respectively, while the European and Americas regions have much lower rates.

Table 1: Estimated TB incidence by WHO region, 2024. Source: WHO Global Tuberculosis Report 2025 [World Health Organization, 2025]. Note: Indonesia was reclassified from SEAR to WPR by WHA Resolution 78 (2025), affecting comparability with earlier editions.

WHO Region	**New cases (thousands)**	**Rate per 100,000**
South-East Asia	3,680	201
Western Pacific	2,910	131
Africa	2,620	207
Eastern Mediterranean	920	112
Americas	350	33
Europe	204	22
Global	**10,700**	**131**

1.2 Twenty-year trend in global incidence

Global TB incidence has been declining since approximately 2000, driven by improved treatment coverage, HIV antiretroviral therapy (ART) scale-up in high-burden African countries, and economic development. The WHO End TB Strategy, adopted in 2015, set a baseline rate of approximately 142 cases per 100,000 and targets a 50% reduction by 2025 and a 90% reduction by 2035.

Table 2 presents global TB incidence at approximately five-year intervals from 2005 to 2024. Values for 2005–2012 are approximate estimates from the WHO Global Tuberculosis Control Reports

published in those years; more recent values are from the 2021 and 2025 editions. All estimates carry considerable uncertainty and historical series are revised retrospectively as new national prevalence survey data become available; absolute case counts before 2015 are particularly subject to upward revision and are not reported here.

Table 2: Global TB incidence rate, selected years 2005–2024. Values marked ≈ are approximate or derived from earlier report editions. Source: WHO Global Tuberculosis Reports, various years [World Health Organization, 2025].

Year	Rate per 100,000	New cases (millions)
2005	≈179	—
2010	≈160	—
2015	≈142	≈10.4
2019	≈130	≈10.0
2020	127	9.9
2022	≈133	≈10.6
2024	131	10.7

The COVID-19 pandemic disrupted TB services sharply in 2020, when newly diagnosed cases fell from 7.1 million in 2019 to 5.8 million in 2020. This reduction in detection is estimated to have increased the pool of undiagnosed, untreated cases and driven a 4.6% rise in estimated incidence between 2020 and 2023 (reaching 10.8 million new cases in 2023, the highest estimate in the modern era). By 2024, improved case-finding had reversed this trend, with incidence falling back to 131 per 100,000 for the first time since the pandemic [World Health Organization, 2025].

Regional trends diverge substantially. Between 2015 and 2024, the European Region achieved the largest proportional reduction (39%), driven primarily by the Russian Federation. The African Region declined 28%, reflecting ART scale-up. Progress was slower in the South-East Asia Region (16%) and the Eastern Mediterranean Region (5.9%). In contrast, the Western Pacific Region and the Americas showed small *increases* of 1.7% and 13% respectively over the same period, the latter driven by sustained upward trends in Brazil and several neighbouring countries.

TB prevalence (active disease at a point in time) exceeds incidence because cases persist for months to years before treatment or self-resolution. WHO estimates global TB prevalence as roughly 1.5–2× the incidence count, implying approximately 15–20 million people living with active TB globally in recent years; however, precise prevalence figures by region require periodic national prevalence surveys and carry wider uncertainty intervals than incidence estimates. Country-level prevalence estimates are published in the WHO Global Tuberculosis Report app and country profiles.

1.3 Natural history in the absence of treatment

Natural history in the absence of treatment is prolonged. A systematic review of pre-chemotherapy era data estimates untreated pulmonary TB has an average duration of about three years and high case fatality, especially in smear-positive disease [Tiemersma et al., 2011]. These estimates provide important anchors for model parameters related to infectious duration and TB-specific mortality.

Recent evidence emphasizes a spectrum of TB disease states, including minimal disease (non-infectious), subclinical disease (infectious but asymptomatic), and clinical disease (symptomatic and infectious) [Richards et al., 2023]. Subclinical TB has epidemiological relevance because it contributes to transmission while escaping symptom-based detection [Kendall et al., 2021]. An in-

dividual participant data meta-analysis of 620,682 adults across 12 nationally representative surveys found that 59% of bacteriologically confirmed TB cases had no persistent cough—rising to 83% after adjustment for false-negative chest radiographs—underscoring how heavily symptom-based detection underestimates the infectious pool [Stuck et al., 2024].

A Bayesian-calibrated model spanning the full infection-to-disease spectrum yields several results that update standard assumptions embedded in compartmental models [Horton et al., 2023]. The model estimates that 92% of newly infected individuals self-clear within 10 years; only 7.9% progress to any form of TB disease, and of those, only 33% reach symptomatic clinical disease. The conventional "5–10% lifetime risk" figure therefore refers specifically to infectious disease and conflates clearance of latent infection with persistent risk. A further finding is that nearly half of those who develop infectious disease undulate between subclinical and clinical states rather than following a unidirectional progression, and only 44% of clinical TB onset occurs within 2 years of infection—compared to 98% of minimal-disease onset—indicating that slow progressors are heterogeneous in ways the simple "fast/slow" latency split does not fully capture.

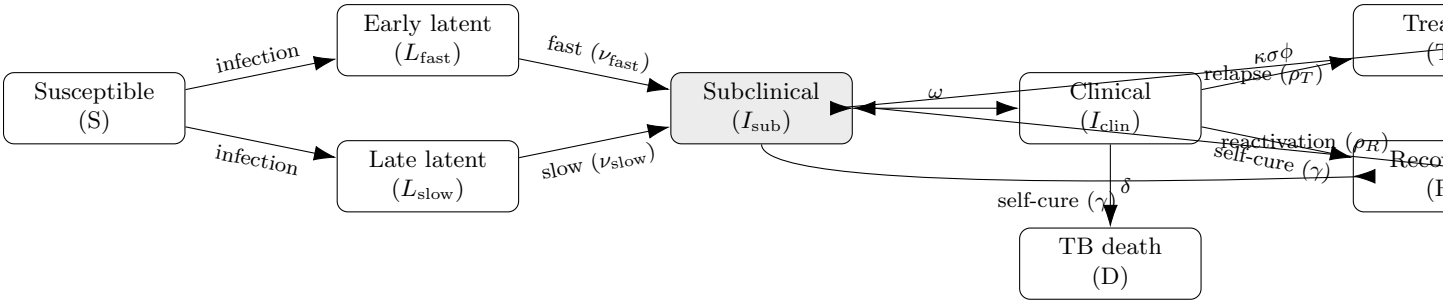

Figure 1: TB natural history with subclinical infectious stage and diagnostic cascade. Shaded compartment (I_{sub}) is not detectable by passive case finding. Detection from I_{clin} requires care-seeking (rate κ), a positive test (probability σ), and linkage to treatment (probability ϕ); the effective detection rate is $\kappa\sigma\phi$.

2 Mathematical modeling approaches for TB

Compartmental ordinary differential equation (ODE) models are a standard entry point for infectious disease modeling. For TB, simple SIR or SEIR models are often insufficient because latency can last years, progression is heterogeneous, and treatment modifies infectious duration. As a result, TB models commonly split the latent period into fast and slow progressors, include explicit treatment compartments, and allow for reactivation and relapse [Vynnycky and Fine, 1997].

Key mathematical tools include the computation of the basic reproduction number R_0 using the next-generation matrix method [Diekmann et al., 1990]. Parameter estimation methods range from least squares to Bayesian inference, and sensitivity analyses (one-at-a-time, PRCC, or tornado plots) are used to identify dominant drivers of transmission dynamics.

Deterministic ODE models provide transparent, interpretable baselines and are well suited for learning. Stochastic or agent-based approaches become more relevant when heterogeneity, household structure, or small population effects dominate the dynamics. This project starts with a deterministic ODE model and provides a clear path toward extensions.

2.1 Stochastic ODE models and uncertainty analysis

Deterministic ODE models suppress variability and mask uncertainty in parameter estimates. Blower et al. [1995] developed a stochastic framework for TB that couples the ODE system with Monte Carlo sampling over uncertain parameters to produce probabilistic projections of epidemic trajectories. A key finding is that TB epidemics evolve over decades to centuries—a timescale set by the slow progression of latent infection and the density of the susceptible pool—so short-horizon projections may be misleading without propagating uncertainty from input parameters. Sensitivity analysis methods such as partial rank correlation coefficients (PRCC) are used to identify which parameters most drive uncertainty in R_0 and incidence projections. This approach remains useful as a complement to deterministic baselines when communicating uncertainty to policy audiences.

2.2 Extended compartmental models with drug resistance and reinfection

More complex ODE models stratify the population further to capture phenomena absent from simple SEIR-like structures. Trauer et al. [2014] constructed a ten-compartment deterministic model calibrated to high-burden settings in the Asia-Pacific. The model adds: (i) a declining hazard of active disease with time since infection, (ii) explicit reinfection during latency and after treatment, (iii) a separate multidrug-resistant (MDR-TB) strain with fitness costs and de novo resistance during treatment, and (iv) a vaccine compartment with temporary efficacy. A critical finding is that the model could not reproduce observed incidence trajectories without allowing reinfection during latency, indicating this mechanism is epidemiologically important in high-burden settings. Additionally, even a modestly fit MDR-TB strain becomes dominant at equilibrium under most parameterizations, illustrating how treatment coverage interacts with drug resistance dynamics in ways that simple two-strain ODE models capture efficiently.

Zwerling et al. [2015] extend this theoretical framework by defining a strain-specific effective reproductive ratio $R_{e,\mathrm{dr}}$ for drug-resistant strains operating within an ongoing drug-sensitive epidemic. A key analytical result is that the rate of *de novo* resistance acquisition during treatment does not enter the expression for $R_{e,\mathrm{dr}}$: only the relative transmission fitness and the relative duration of infectiousness of the resistant strain determine its long-run competitive advantage. This has a counterintuitive surveillance implication: DR-TB prevalence in the first $\sim$5 years following launch of a new first-line drug regimen is dominated by acquisition-driven dynamics, and may therefore not predict the long-term trajectory, which is governed by transmission fitness. Drug susceptibility testing (DST) deployed alongside new regimens reduces the duration of DR-TB infection and hence lowers $R_{e,\mathrm{dr}}$, giving it outsized long-run benefit relative to what short-term surveillance data alone would suggest.

2.3 Individual-based and microsimulation models

When population heterogeneity is a first-order concern—for example, when modeling targeted interventions in low-incidence settings—individual-based or microsimulation models offer advantages that ODE compartmental models cannot easily replicate. Goodell et al. [2019] developed a discrete-time stochastic microsimulation for California in which each simulated individual carries attributes including age, race/ethnicity, nativity, and medical risk factors (HIV, diabetes, immunosuppressive therapy, among others) that modify their reactivation rate. The model captures semi-random mixing by nativity group and is calibrated to historical case counts from 2001 to 2014. Compared to population-level ODE projections, the individual-based structure allows the authors to estimate cost per QALY for interventions targeted at specific risk groups, and to model the threshold scale-up

needed to hit the WHO pre-elimination target of < 10 cases per million. The trade-off is substantially higher computational cost and greater data requirements for demographic parameterization.

A 2024 systematic review of ABMs for TB transmission [Bui et al., 2024] provides the most comprehensive survey of the field to date, covering 26 studies published through December 2022 across high-, moderate-, and low-burden settings. The review confirms that ABMs excel at capturing heterogeneous mixing, household structure, and high-risk transmission clusters—features that ODE compartmental models cannot replicate without strong aggregation assumptions. Most included studies incorporated population heterogeneity and household structure, but spatial structure and economic evaluation modules were commonly absent. A reproducibility gap is also evident: only 8 of 26 studies shared code openly. The authors recommend open-source code sharing and standardised reporting as immediate priorities for the field. A practical implication is that ABMs and ODE models should be seen as complementary: ODE models remain the appropriate baseline for aggregate transmission dynamics and policy projections at national scale, while ABMs are better suited to questions dominated by network structure, targeted interventions, or elimination thresholds where individual heterogeneity is first-order.

2.4 Risk-factor stratified models: nutrition and biomarkers

A second class of model extensions stratifies the population along a continuous biological risk factor—most commonly body mass index (BMI) or glycaemic status—rather than by transmission or drug-resistance status. The key modelling step is to replace scalar progression and treatment-outcome rates with values that depend on the stratum, allowing the force of infection on downstream compartments to vary by nutritional or metabolic state.

Nutrition and BMI. Oxlade et al. [2015] constructed a compartmental transmission model for the Central Eastern states of India in which the population is stratified by BMI. Progression rates from latent to active TB are assigned per stratum following empirical estimates of the BMI–TB risk relationship, and strata exchange members as nutritional status changes over time. Simulations showed that even a modest reduction in undernutrition could avert 4.8 million TB cases and 1.6 million TB deaths over 20 years; complete elimination of undernutrition was projected to avert 9.4 million cases, underscoring how deeply TB burden in high-burden settings is driven by nutritional risk. Building on this framework, Mandal et al. [2024] applied a comparable deterministic compartmental model to the WHO South-East Asia Region, explicitly stratifying individuals as well-nourished or undernourished and coupling the model to country-specific undernutrition prevalence estimates. The study simulated two intervention types—nutritional rehabilitation of TB patient contacts versus a population-wide programme—and found that the population-wide approach reduces cumulative incidence 2–3 fold more than the targeted contact strategy; in India specifically, rehabilitating 30% of the undernourished population annually would avert 15.9% of cumulative TB incidence between 2023 and 2030.

A more recent extension couples the TB transmission model to an external food-system model [Clark et al., 2025]. The TB component uses four explicit BMI strata (moderate–severe thinness, mild thinness, normal, overweight), with stratum-specific progression and treatment-outcome rates. The LandSyMM food-system model projects how supply-chain shocks—parameterised as the stylised energy-price and export-restriction scenarios arising from the war in Ukraine—shift the population-level BMI distribution, which is then fed forward into the TB model. In a worst-case scenario, the combined shock increases TB incidence by 5.0% and mortality by 4.9% in India by 2035, equivalent to an additional 1.1 million TB episodes. The approach illustrates how global

macroeconomic shocks can propagate through nutritional pathways into infectious disease burden, and provides a template for coupling TB models to broader health determinant models.

Diabetes as a biomarker-mediated risk factor. Diabetes mellitus (DM) substantially increases the risk of TB and is increasingly prevalent in high-burden settings. Awad et al. [2019] developed an ODE model stratified by DM status, calibrated to India, and used it to map ten plausible biological pathways by which DM modifies TB natural history and treatment outcomes—including increased susceptibility, accelerated fast progression, higher smear-positivity, elevated infectiousness (through higher bacterial load), increased TB mortality, and reduced treatment success. Using a standardised effect size for each pathway, the model quantifies both direct effects (captured by hazard ratios comparing DM to non-DM individuals) and indirect effects arising from onward transmission through the expanded infectious pool. A notable finding is that pathways that have no impact on the individual-level hazard ratio, such as increased infectiousness, can still account for up to 30% of the population-attributable fraction through amplified transmission—a result inaccessible to conventional epidemiological designs. This work demonstrates that biomarker-stratified ODE models can reveal mechanistic contributions to population-level TB burden that observational hazard-ratio analyses systematically miss.

Collectively, these risk-factor stratified models share a structural logic: the standard compartmental states (S, L, I, T, R) are replicated across risk strata, stratum-specific rates substitute for scalar parameters, and movement between strata represents changes in the underlying biomarker. The design choice of how many strata to use, and how to parameterise inter-stratum transition rates from survey data, is a key modelling decision with direct implications for policy inference.

2.5 Models of diagnostic interventions and the transmission curve

Compartmental models can be coupled with explicit representations of the diagnostic and care-seeking pathway to evaluate the population-level impact of novel diagnostic tests. Zwerling et al. [2015] synthesise this literature and provide a unified conceptual framework centred on the *transmission curve*: the time-varying rate at which an index case generates secondary infections over the course of their infectious episode. The shape of this curve—which depends on bacterial shedding dynamics, the rate of contact with susceptible individuals, and host susceptibility—determines how much of the total transmission burden can realistically be averted by a diagnostic intervention deployed at a given stage of disease.

A critical implication is that passive case finding (relying on symptomatic presentation) is structurally late relative to the transmission peak. Models that assume zero infectiousness prior to symptom onset systematically overestimate the impact of passive diagnostic strategies by 50% or more, because they miss transmission occurring during the subclinical phase. Active case detection—identifying prevalent cases in the community irrespective of symptom status—accesses the earlier, steeper portion of the transmission curve and delivers greater reductions in R_e per case detected. An additional constraint on passive diagnosis is contact saturation: the pool of susceptible household contacts often reaches saturation before an index case is diagnosed, and modelling suggests that 75–95% of household infections may already have occurred by the time contact tracing is initiated, limiting the transformative potential of household-tracing programs in isolation.

Smear-negative cases, which escape traditional sputum-smear detection, can still contribute a substantial fraction of population-level transmission; models must represent this group explicitly to give unbiased projections of the impact of microbiological diagnostics. More broadly, the effectiveness of any diagnostic intervention is not solely a function of test sensitivity: it also depends on access to care, the diagnostic algorithm, linkage to treatment, and the epidemiological context.

These structural considerations complement the cost-effectiveness analyses in Section 4 and explain why identical tests may have very different population-level impact across settings with different care-seeking patterns.

3 Interactions with HIV

HIV co-infection is the single most important biological modifier of TB natural history at the population level. HIV-positive individuals have a dramatically elevated risk of progressing from latent TB infection to active disease—estimated at 20–30 times the risk in HIV-negative individuals—and this effect intensifies as $CD4^+$ cell counts decline [Getahun et al., 2010, Lawn and Zumla, 2011]. The interaction is bidirectional: TB accelerates HIV disease progression and is the leading cause of death among people living with HIV globally, while HIV reshapes TB epidemiology by expanding the infectious pool and shortening the interval between infection and disease [Corbett et al., 2003]. In 2024, approximately 600,000 new TB cases occurred among HIV-positive individuals, and HIV-associated TB accounted for a disproportionate share of TB deaths, particularly in sub-Saharan Africa [World Health Organization, 2025].

3.1 Biological mechanisms and parameter modifications

HIV modifies several parameters in standard TB compartmental models. The most important effects are:

- **Increased progression rate.** HIV infection raises both the fast-progression rate (ν_{fast}) and the slow-reactivation rate (ν_{slow}), with the magnitude depending on $CD4^+$ count. Models that stratify by HIV status typically multiply the baseline progression rates by a CD4-dependent relative risk ranging from $\sim$3 at high CD4 counts ($>$ 500 cells/μL) to $>$15 at counts below 200 cells/μL [Williams and Dye, 2003].
- **Increased susceptibility to (re)infection.** HIV-positive individuals who clear a primary TB infection remain at elevated risk of reinfection, further compounding the progression effect.
- **Modified disease presentation.** HIV-associated TB is more often smear-negative and extrapulmonary, reducing sensitivity of sputum-smear-based diagnostics and delaying detection. In the subclinical/clinical framework of Section 1, this can be represented as a lower care-seeking rate κ or lower test sensitivity σ for HIV-positive individuals.
- **Higher case fatality.** TB-specific mortality (δ) is substantially elevated in HIV co-infected individuals, and untreated TB in turn accelerates progression to AIDS.

3.2 Modeling approaches for TB–HIV co-dynamics

The simplest approach replicates the TB compartmental structure across HIV strata, yielding a "crossed" model in which each TB state (S, L_{fast}, L_{slow}, I_{sub}, I_{clin}, T, R) exists for HIV-negative and HIV-positive sub-populations, with HIV-status-dependent rates. This doubles (or further multiplies, if CD4 strata are used) the number of state variables but preserves the ODE framework.

Bacaër et al. [2008] developed a six-compartment model for the joint TB and HIV epidemics in a South African township where HIV prevalence exceeded 20% and TB notification rates approached 2,000 per 100,000 per year. The model tracks the two epidemics as coupled dynamical

systems sharing a common population, and evaluates the relative effectiveness of condom promotion, increased TB case detection, TB preventive therapy, and antiretroviral therapy (ART). A key structural finding is that the coupling between the two epidemics makes it impossible to project TB trajectories accurately without simultaneously modeling the HIV epidemic: TB incidence is a function of HIV prevalence, which is itself dynamic.

Sharomi et al. [2008] provided a rigorous analytical treatment of HIV/TB co-dynamics, demonstrating that the TB-only sub-model exhibits backward bifurcation—meaning a stable disease-free equilibrium can coexist with a stable endemic equilibrium even when the reproduction number $R_0 < 1$. This has a practical implication for control: reducing R_0 below unity is necessary but may not be sufficient to eliminate TB in populations with HIV co-infection, because the co-infection dynamics sustain a second stable equilibrium that standard threshold arguments do not predict. The analysis also found that HIV-only treatment strategies (ART) averted more co-infection cases than TB-only strategies when resources were limited, highlighting the cross-disease benefit of ART.

3.3 Impact of antiretroviral therapy on TB dynamics

ART reduces TB risk in HIV-positive individuals by restoring immune function, but the magnitude of the population-level effect depends on coverage, the CD4 threshold for ART initiation, and the timing of scale-up relative to the maturity of the HIV epidemic. Williams and Dye [2003] used a compartmental model to show that initiating ART at 200 $CD4^+$ cells/μL with 90% coverage would prevent no more than 20% of HIV-associated TB cases, whereas initiation at 500 cells/μL could prevent 40–60%. This result arises because much of the excess TB risk accumulates at moderate immunosuppression (CD4 200–500), a range that late-initiation strategies leave unprotected.

Pretorius et al. [2014] compared three independent TB–HIV transmission models for South Africa and found that expanding ART eligibility to all HIV-positive individuals was projected to reduce cumulative TB incidence by 6–30% over 2014–2033, with an estimated one TB case averted for every 10–13 additional person-years of ART. The between-model variation in this estimate reflects structural uncertainty in how ART modifies infectiousness and in the assumed rate of immune reconstitution, underscoring the value of multi-model comparisons for policy-relevant projections.

3.4 Implications for model design

For models intended to describe settings with substantial HIV prevalence, ignoring the TB–HIV interaction leads to biased projections of both incidence trends and intervention impact. The minimum extension required is a two-stratum (HIV−/HIV+) replication of the TB compartmental structure with stratum-specific progression, mortality, and diagnostic parameters. More refined models add CD4-count strata, explicit ART compartments with time-on-treatment effects, and dynamic HIV transmission. The structural trade-off is the same as for other risk-factor stratified models (Section 2.4): additional strata improve biological realism at the cost of more parameters, and the appropriate level of complexity depends on the policy question and the data available for calibration. In low-HIV-burden settings, the interaction can be safely neglected or handled through an attributable-fraction adjustment without explicit HIV strata.

4 Economic evaluation approaches for TB

Economic evaluation methods are used in TB research to assess whether interventions deliver sufficient health benefit relative to their costs, and to prioritize limited resources across competing strategies. The three most common frameworks are cost-effectiveness analysis (CEA), cost-utility

analysis (CUA), and budget impact analysis (BIA). CEA and CUA typically compute an incremental cost-effectiveness ratio (ICER):

$$\text{ICER} = \frac{\Delta\text{Cost}}{\Delta\text{Health outcome}},$$

where the health outcome is expressed in natural units (e.g., cases averted, deaths averted) for CEA, or in quality-adjusted life years (QALYs) or disability-adjusted life years (DALYs) for CUA. The WHO convention expresses TB burden in DALYs, making cost-per-DALY-averted the dominant metric in TB intervention evaluations. An intervention is conventionally deemed cost-effective if its ICER falls below one to three times the per-capita GDP of the country under study, though this threshold is contested.

A distinctive feature of TB economic evaluations is that they frequently integrate the economic analysis directly with a transmission model. Because TB interventions such as improved diagnostics or treatment completion reduce transmission, a static model that ignores downstream prevented infections will underestimate the benefit. Menzies et al. [2012] demonstrate this approach: they coupled a calibrated dynamic transmission model for five southern African countries with a cost module to evaluate Xpert MTB/RIF versus sputum-smear diagnosis. Over a ten-year horizon, the intervention was projected to avert 132,000 TB cases and 182,000 deaths at a cost-effectiveness of approximately $959 per DALY averted—within the cost-effectiveness threshold for most of the study countries. Importantly, a large fraction of the additional costs arose from antiretroviral therapy for HIV-positive individuals who survived due to better TB case-finding, illustrating how transmission-model-based economic evaluations can reveal unexpected cost drivers through downstream effects.

Budget impact analysis differs from CEA in that it asks not whether an intervention is worth its cost, but whether the health system can afford it. For TB, scale-up of MDR-TB treatment or novel diagnostics often passes cost-effectiveness thresholds yet fails budget impact analyses because the upfront financial requirement exceeds available program budgets. Taken together, CEA/CUA and BIA serve complementary roles: the former informs whether to adopt an intervention, and the latter informs whether and how quickly the health system can realistically do so.

5 The SELT(R) model used in this project

The toy model implemented here is a discrete-time stochastic microsimulation with seven state variables, chosen to reflect the biological features described in Section 1 together with two extensions motivated by Section 1: (i) an explicit subclinical infectious stage informed by the disease spectrum literature [Richards et al., 2023, Kendall et al., 2021], and (ii) a disaggregated diagnostic cascade following the transmission-curve framework of Zwerling et al. [2015].

5.1 Compartments and biological rationale

- **Susceptible (S)** – individuals with no prior infection, fully at risk.
- **Early latent (L_{fast})** – recently infected fast progressors (∼10% of new infections), who develop active TB within months.
- **Late latent (L_{slow})** – recently infected slow progressors (∼90%), who carry a low but persistent long-term reactivation risk.
- **Subclinical infectious (I_{sub})** – active TB in the pre-symptomatic or minimally symptomatic phase; capable of transmission at reduced intensity (relative infectiousness $\varepsilon < 1$) but not detectable by passive symptom-based case finding.

- **Clinical infectious (I_{clin})** – symptomatic active TB; full infectiousness; accessible to passive diagnosis. The diagnostic pathway from I_{clin} to treatment involves three sequential steps: care-seeking at rate κ, a positive test result with probability σ, and linkage to treatment with probability ϕ, yielding an effective detection rate $\kappa\sigma\phi$.
- **Treated (T)** – individuals on treatment; no longer contributing to transmission, but subject to relapse into I_{sub}.
- **Recovered (R)** – self-cured from either infectious stage; no longer infectious, but reactivation into I_{sub} is possible.

5.2 Force of infection and the transmission curve

The force of infection is:

$$\lambda = \frac{\beta\,(\varepsilon\, I_{\text{sub}} + I_{\text{clin}})}{N}$$

where $\varepsilon \in (0, 1)$ captures the partial contribution of subclinical cases to transmission. Setting $\varepsilon = 0$ recovers a model in which only symptomatic cases transmit; setting $\varepsilon = 1$ recovers the original single-compartment model. The parameter ε operationalizes the transmission-curve concept of Zwerling et al. [2015]: passive diagnosis arrives late relative to the infectiousness peak, and the proportion of transmission occurring before symptom onset (indexed by ε) determines how much passive case-finding can actually avert.

5.3 Basic reproduction number

With two infectious stages, R_0 is derived from the next-generation matrix:

$$R_0 = p_I \cdot \beta \cdot \frac{\varepsilon\, d_{\text{clin}} + \omega}{d_{\text{sub}} \cdot d_{\text{clin}}}$$

where $d_{\text{sub}} = \omega + \gamma + \mu$, $d_{\text{clin}} = \kappa\sigma\phi + \gamma + \delta + \mu$, and

$$p_I = f \cdot \frac{\nu_{\text{fast}}}{\nu_{\text{fast}} + \mu} + (1 - f) \cdot \frac{\nu_{\text{slow}}}{\nu_{\text{slow}} + \mu}$$

is the probability a new latent infection eventually reaches I_{sub}. Improving test sensitivity σ or linkage ϕ increases d_{clin}, shortening the mean clinical infectious duration and thereby reducing R_0.

The flow diagram (Figure 1) summarizes all compartments and transitions. The full ODE system, R_0 expression, and default parameter values with literature sources are documented in notebook `01_tb_model.ipynb`.

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